



SADAKVISION

Sensor–vision fusion for scalable smartphone
road monitoring

INDIS 2026

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INDIA, 2024

2,385

lives lost in pothole-related crashes

5,432

crashes

4,643

injured

On 6.7 million km of road, inspection still means a manual survey, or a LiDAR rig costing five figures (USD) per vehicle.

Sources: Lok Sabha Unstarred Q. 2175, 12 Feb 2026; RIEGL USA (2026)

SadakVision · INDIS 2026 02

A phone can **feel** a pothole or **see** one, but not both, and not for long

APPROACH 1

Sensor only

Reacts to every vertical shock, including unmarked speed humps.

Precision 0.48

APPROACH 2

Vision only

Finds visible damage but cannot measure impact; dark patches cause false alarms.

No severity estimate

APPROACH 3

Always-on video

Runs the detector on every frame; the phone overheats and throttles.

Throttles after 32 min

DESIGN QUESTION

Can a phone look only when it has felt something?

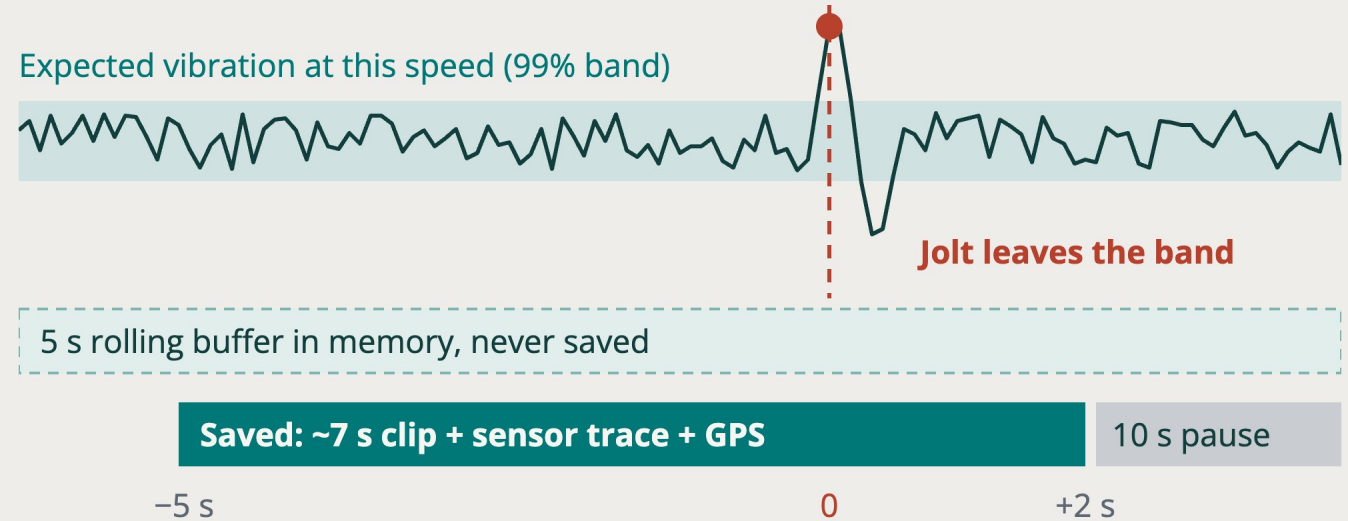
FEEL FIRST, THEN LOOK

NOVELTY

The motion sensor does not classify potholes. It decides when the camera's buffer is kept.

The phone learns what normal vibration feels like at its current speed. Only a jolt outside that band saves video.

One trigger, second by second (schematic)



55 m

of road already in the buffer at 40 km/h

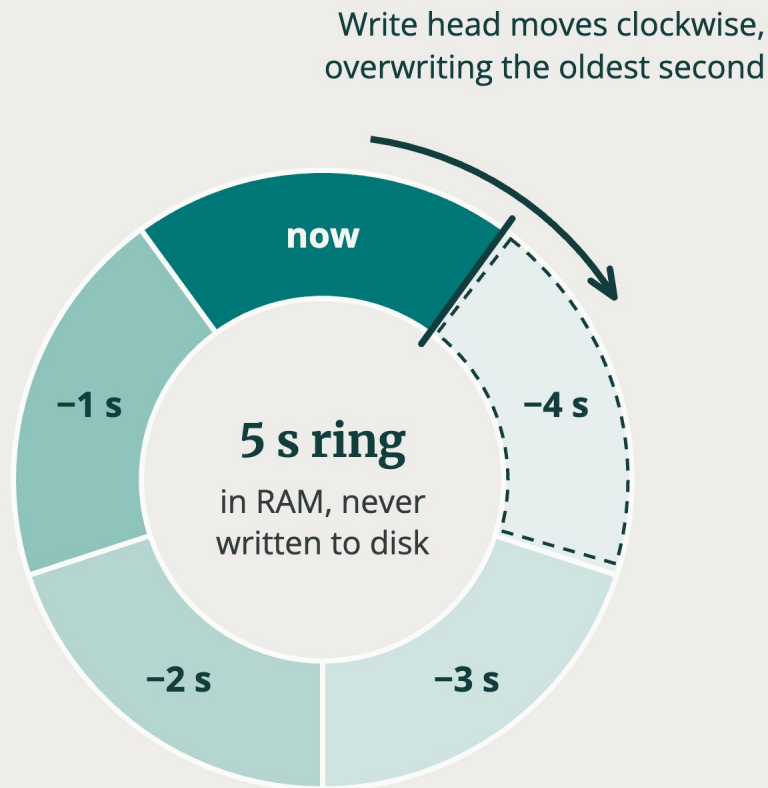
0.5 m/s²

beyond the band triggers a save

100 Hz

accelerometer + gyroscope, with GPS speed

| Novelty: a **sensor-triggered circular buffer**



WHEN THE SENSOR TRIGGERS

Last 5 s, flushed from the ring

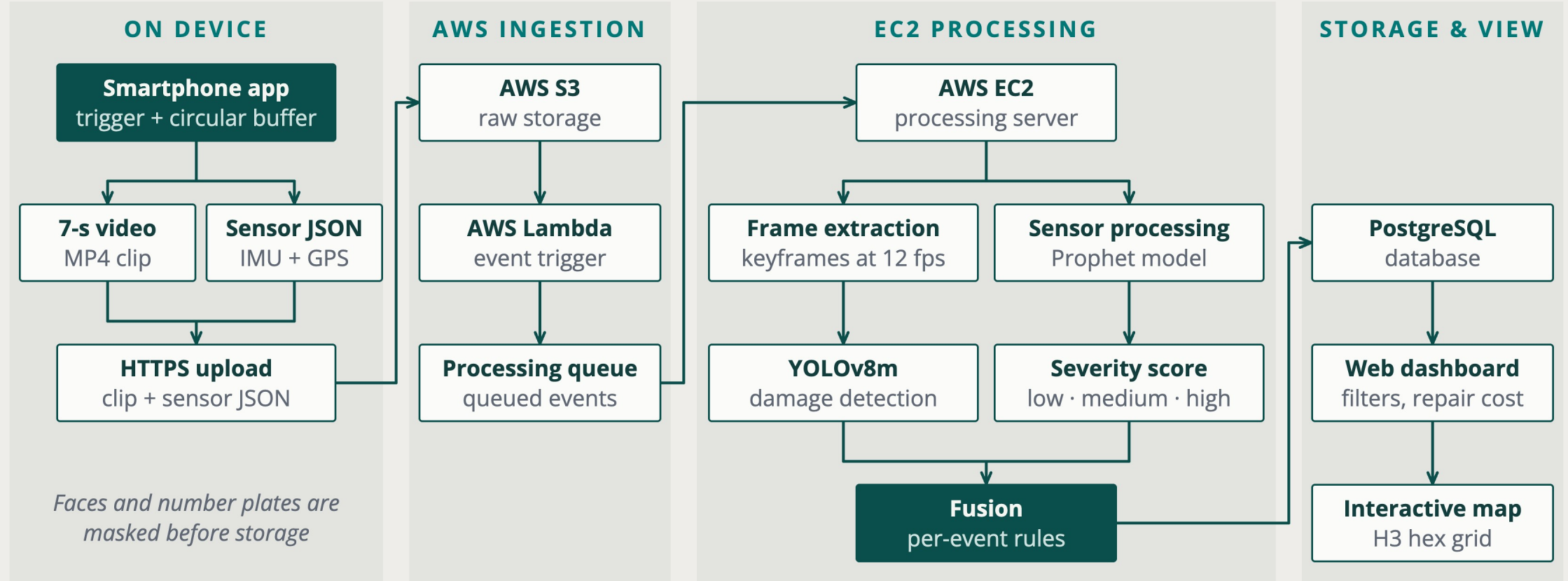
+ ~2 s in progress

= one ~7 s MP4 with its **sensor trace and GPS**, saved as a single event. Then a 10 s pause.

Table 2. How the motion sensor is used

System	Role of the motion sensor	Video
Nericell (2008)	Final classifier	No video
Pothole Patrol (2008)	Final classifier	No video
Ramesh et al. (2022)	Separate model; fused later by clustering	Not paired
SadakVision	Trigger that keeps a pre-roll video buffer	Paired per event

System architecture: **from phone to map**



■ **Our contributions:** sensor-triggered circular buffer on the phone (C1) and per-event sensor-vision fusion (C2)

Two users: the **driver** who shouldn't notice it, the **engineer** who must act on it

FOR THE DRIVER

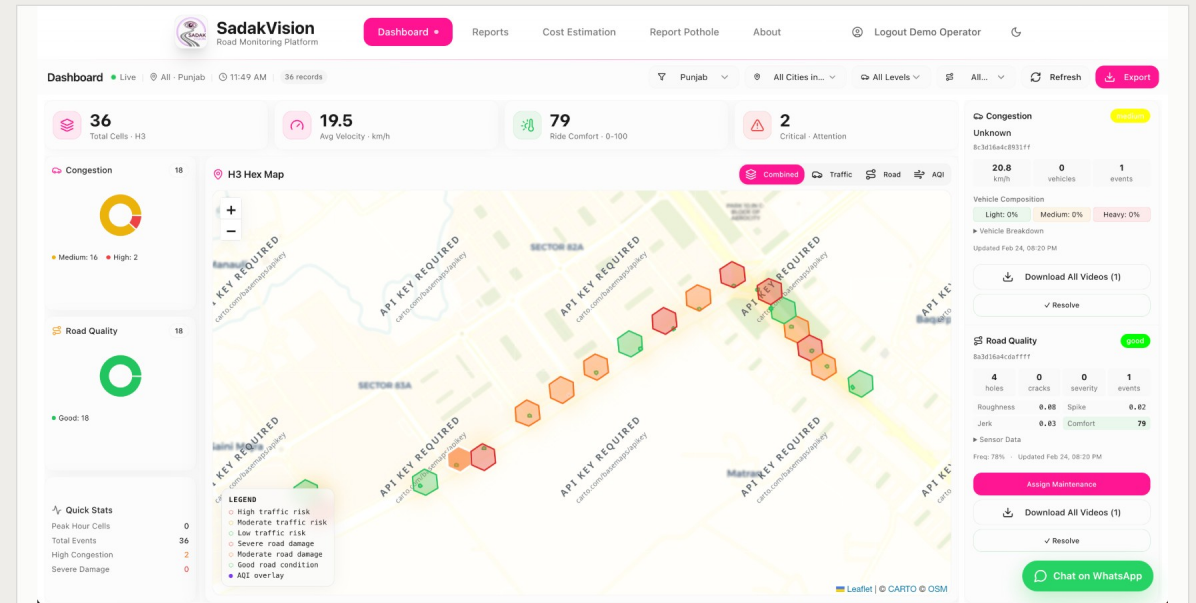


One tap to start, then silent: no alerts, nothing to watch

Mirror mount keeps the windshield clear

3-vehicle trial: drivers reported no distraction

FOR THE MUNICIPAL ENGINEER



Colour-coded severity

Repair-cost estimate

Assign repairs

Captured on **Chandigarh–Punjab roads**



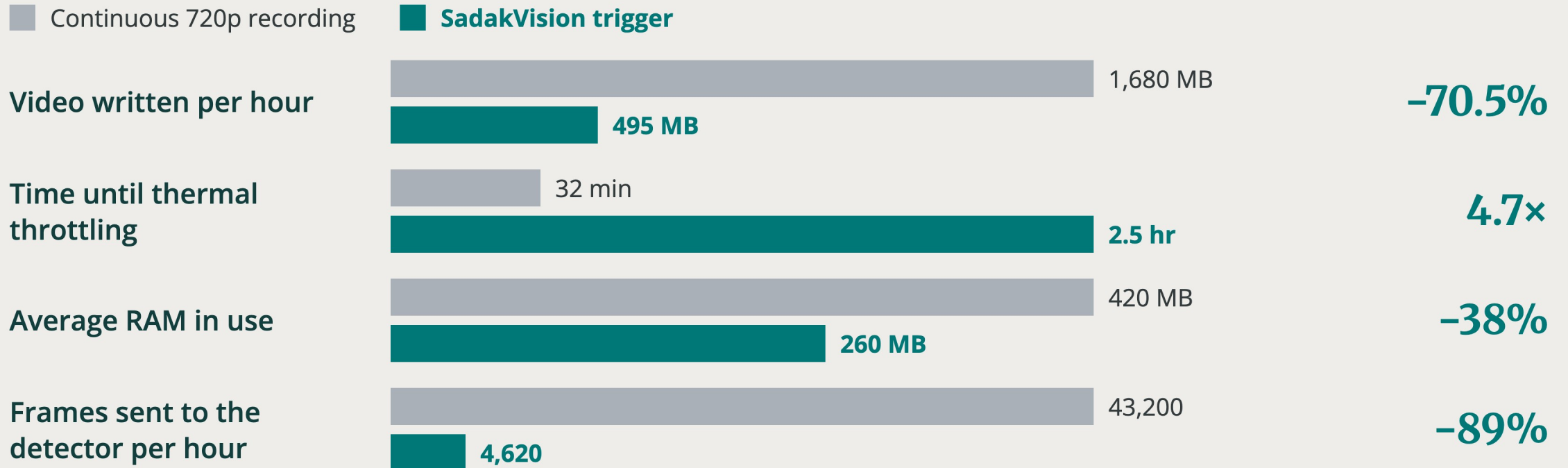
40 days of daytime, dry-weather driving

3 four-wheelers, phone on the mirror

27,609 images, 18,496 with potholes

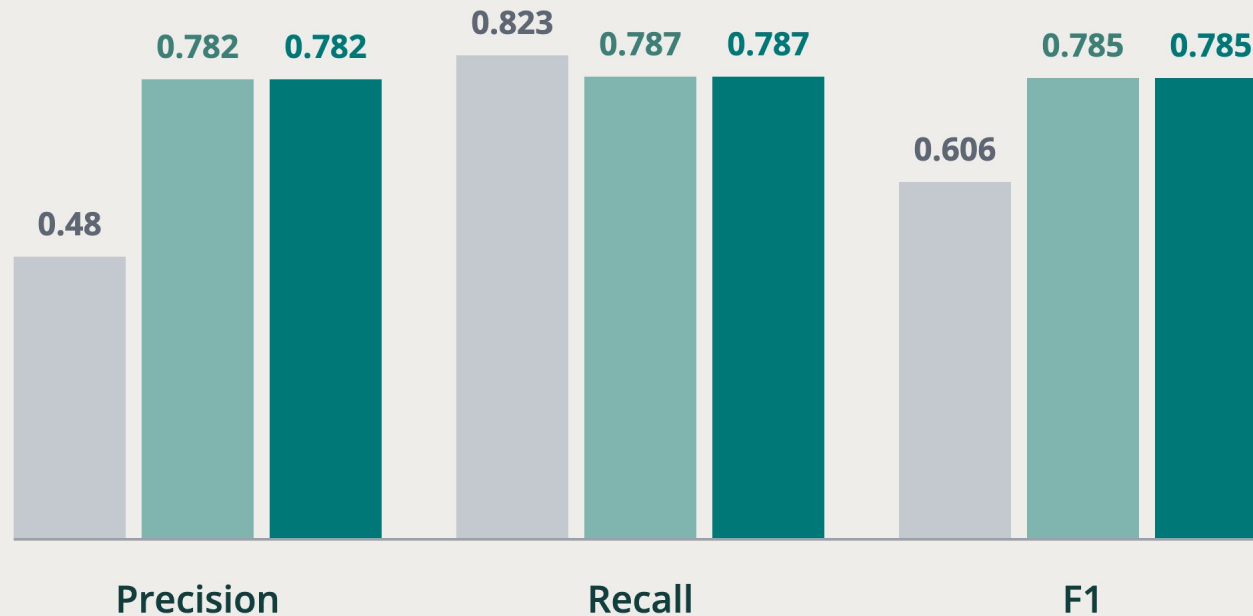
70% fewer writes, 4.7× longer before the phone overheats

Effect of the sensor-triggered circular buffer (C1)



Vision-level accuracy, **89% less compute**, plus severity

■ Sensor only ■ Vision only ■ Fusion (SadakVision)



Same F1

as vision-only, by design: the sensor decides what the camera looks at, vision gives the verdict.

-89%

frames sent to the detector: 4,620 vs 43,200 an hour.

+ Severity

low / medium / high from the sensor trace.
Checked so far on 62 human-labelled events.

| Where it stops today, **and what's next**

LIMITS TODAY

Daytime, dry weather only (11 am – 5 pm)

Four-wheelers only

One region: Chandigarh–Punjab

Severity checked on only 62 labelled events

Usability tested informally, not yet with SUS

NEXT

Two-wheelers: new triggers and severity maps

Adaptive thresholds for long, smooth stretches

On-device inference at the edge

IRI-based maintenance planning

Night, fog and rain data; learned fusion



Thank You

Sense first, see second.



sadakvision.com

App, dashboard and reports are live



Dixon IoT Lab · Plaksha University

Backup · The trigger and fusion rules

ON THE PHONE (100 Hz)

$$Z(t) = \text{trend}(t) + \beta_v \cdot v(t) + \varepsilon(t)$$

Prophet: vertical acceleration with GPS speed $v(t)$ as regressor. 99% interval, changepoint prior 0.15, no seasonality.

Anomaly if $Z(t) < \hat{Z}_{\text{low}}$ or $Z(t) > \hat{Z}_{\text{high}}$

Group consecutive anomalies. Trigger if $D_{\text{min}} < -0.5 \text{ m/s}^2$ or $D_{\text{max}} > 0.5 \text{ m/s}^2$. Then flush the ~5 s buffer (clip $\approx 7 \text{ s}$), save the JSON trace and GPS, and pause 10 s.

IN THE CLOUD (PER EVENT)

Report if $c \geq 0.45$ and $S \geq \tau$

Report if $c \geq 0.75$

Review if $S \geq \tau$ and $c < 0.45$

c = YOLOv8m confidence on frames at 12 fps. S , τ = sensor anomaly score and its trigger threshold (same model as the trigger).

Severity bands (low / medium / high) come from thresholds on D_{min} / D_{max} , not from a trained classifier.

| Backup · YOLOv8m, per class (test set)

Class	Precision	Recall	mAP50	mAP50-95
Crack	0.897	0.920	0.895	0.663
Garbage	0.694	0.475	0.444	0.207
Manhole	0.829	0.871	0.865	0.576
Pothole	0.782	0.787	0.763	0.469

Mean over classes: mAP50 0.742 · mAP50-95 0.479

Garbage is the weakest class; its imagery comes from public sources.

TRAINING

YOLOv8m, COCO-pretrained, 640×640

44 epochs (early stop), batch 32, LR 0.01

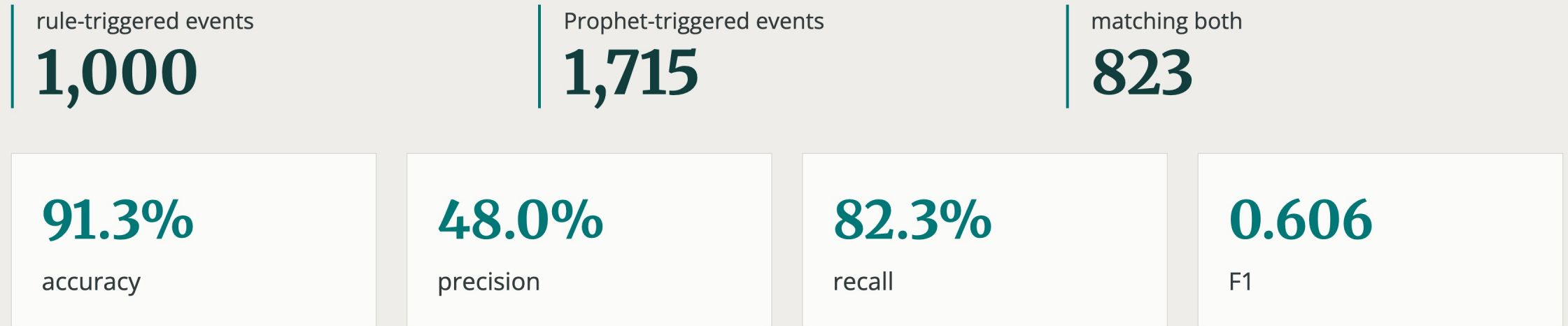
90 / 5 / 5 split **by driving route**, so near-duplicate frames never leak across sets

Flip and rotate augmentation

Validation mAP50: 75.6%

| Backup · **Prophet trigger vs the old g-force rule**

Compared with the previously deployed rule (vertical g-force > 4), over 12,260 sensor samples.



High recall, more false positives: the trigger over-collects, and the vision stage filters the extras out.

Backup · How it sits against prior work

Method	Modality	Reported score	Severity?
Ramesh et al. 2022 (YOLOv5)	Vision	87.5% mAP	No
Ramesh et al. 2022 (LSTM)	Sensor	94% acc.	Depth estimate
Doshi & Yilmaz 2020	Vision	F1 0.628	No
Varona et al. 2020	Sensor	93%	No
SadakVision	Fusion	76.3% pothole mAP50	Yes: RMS acceleration

Not directly comparable: each study uses its own data, classes and protocol. Our difference is **where the sensor sits**: it triggers the camera instead of being the final classifier.